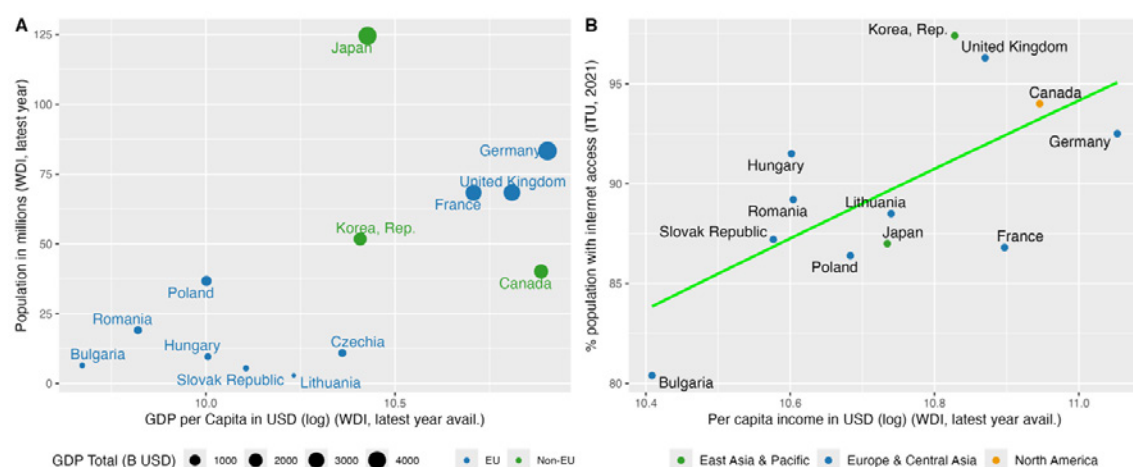


► 2 Tasks and occupations in the 6-digit system in Poland

Within Europe, Poland is quite representative of the larger group of Central and Eastern European (CEE) countries, albeit with a significantly larger population (38 million) than most of its peers. Globally, the CEE ranks between the high-income advanced economies and the emerging economies of the Global South. With respect to access to the internet, Poland ranks below UK and Germany, but higher than Japan and France. Thus, for assessing the potential of task automation, it seems reasonable to assume that Poland is representative of the upper threshold of automation potential, due to high availability of digital equipment and widespread internet access.

► Figure 1. Income- and population-based similarities (A) and access to the internet (B) across countries



The Ministry of Family, Labour and Social Policy (MOL) in Poland maintains a system of occupational classifications, with task and job descriptions that are regularly updated.² The system is used by Statistics Poland (GUS) and other state institutions for labor market analysis. At the most detailed, 6-digit level, this framework contains 2541 occupations and 29,753 corresponding tasks (Table 1). Since the system is publicly available online, we initially designed a scraping bot that crawled through individual occupational pages and downloaded job descriptions and corresponding tasks into a structured data frame.³ The extracted content was subsequently cross verified against a tabular occupational classification provided by the Ministry. The Polish classification system is compatible with the hierarchical structure of ISCO-08, which means that we could easily append an additional mapping of occupations from 1 to 4-digit level (Table 1).

² The occupational classification database (KZIS) was first published in 2014 and has been regularly updated, with the latest validation on 1 January 2025. The classification was introduced by the Regulation of the Minister of Labour and Social Policy on the classification of occupations and specializations for labor market purposes and its scope of application, dated 7 August 2014 (Dz.U. z 2018 r. poz. 227, z 2021 r. poz. 2285, z 2022 r. poz. 853, Dz.U. z 2024 r. poz. 1372). Additionally, the Ministry publishes a mapping to ISCO-08. See: <https://psz.praca.gov.pl/rynek-pracy/bazy-danych/klasyfikacja-zawodow-i-specjalnosci>.

³ Web scraping was primarily conducted to demonstrate the potential of acquiring non-traditional data from publicly accessible sources, in contexts where such activity is not restricted by server settings or legal regulations. The process adhered to responsible data collection practices, including appropriate frequency settings to avoid overloading servers.

We stress that these classifications merely suggest the general trends and types of occupations at different levels of exposure, since any type of task scoring system carries a degree of subjectivity. While the abilities of GenAI and its derivative technologies evolve rapidly, many additional local factors, not covered in this global study, can either speed up and expand, or limit the impact of these technologies on specific occupations. The advantage of this more detailed categorization of occupations into four progressive gradients is that it will allow for tracking the movement of occupations over time, and for occupational-level comparisons across national classification systems with varying collections of tasks.¹⁹

¹⁹ To address the weakness of missing country-level nuance in this global index, we develop an AI-based predictor described in Section 8, which captures the general scoring structure and relative score differences across 2,861 tasks in our database. We can apply this knowledge repository to predict scores for more specific tasks in national-level classification systems. For demonstrative purposes, we intend to apply this method to the context of 6-digit occupations in Poland (Gmyrek et al., forthcoming in 2025), with the possibility of replicating the method for other countries where detailed occupational descriptions are available.

► Conclusion

In this paper, we have presented the methodology used to develop a new global index of occupational exposure to GenAI, as well as new estimates of the global effects on employment. We build on the approach developed by the ILO in 2023, but with several notable refinements. First, by beginning our analysis with the Polish 6-digit occupational structure, we expand by ten-fold the amount of detailed task information available for the analysis of task automation. In addition, using a representative sample from the 29,753 tasks in the Polish occupational classification system and a survey of 1,640 people in Poland, we captured the opinions of each 1-digit ISCO-08 group on the potential for task automation in the job categories in which they are currently employed (52,558 data points). We then compared this input with an expert survey and conducted subsequent rounds of revisions among a smaller international expert group for the tasks with the highest degree of disagreement. Based on this process, we built a repository of knowledge with adjusted task scores that goes beyond Poland's national specificities and reflects a broader consensus on the task automation potential of current GenAI tools.

We subsequently used this repository to develop an AI assistant that could predict scores for these tasks based on this additional information as well as the technical documentation of ISCO-08. These scores update the original set of scores produced by the ILO in 2023 (Gmyrek et al., 2023). The use of a verified "knowledge base" presents an important methodological improvement, which greatly enhances stability of our predictions across independent LLMs (see Section 6), and which can be used to produce similar score predictions for other occupational classification systems that contain job-specific tasks. In addition, we make all task-level scores publicly available, which also makes it possible for researchers to organize the occupational exposure in different ways than our proposed framework of exposure gradients.

The new global index presented in this paper is generic, due to the nature of ISCO-08 and the structure of tasks associated with occupations in this framework. Its definite strength – similar to ISCO-08 – is that it provides a common global denominator and a direct link to national labour forces surveys held in the ILO harmonized micro data repository, which relies on the same classification of occupational structures. Even though all countries do not use ISCO-08, mappings of the national systems to this common international standard are available. This eliminates the need for crosswalks from other national systems (e.g. O*NET), which involves tenuous assumptions about the similarity of detailed occupations in the technologically advanced context of the USA to other countries, especially non-HICs. The simple and comparable character of our ISCO-08 based index enables direct production of global, regional and income-based employment estimates, based on the same statistical modelling techniques that are used by the ILO for other global figures (see Section 11).

Our 2025 scores are presented in a revised occupational classification system, adopting a framework of four progressively increasing exposure gradients, that better highlights the importance of potential AI exposure and task variability for determining possible outcomes. We produced a new set of global estimates of GenAI exposure based on this revised classification.

The new index maintains many of the headline findings of the 2023 methodology. Clerical occupations continue to have the highest exposure levels, albeit with some important downward revisions to individual tasks scores. Such occupations include data entry clerks, typists, accounting, bookkeeping clerks and administrative secretaries. Additionally, some occupations – such as financial analysts, and web and multimedia developers (gradient 4), as well as bank tellers, translators, applications programmers, investment advisers (gradient 3) – have increased their